Technical Report for Road Extraction ArcGis Integration

2026-10-20

Contents

1 Preambule **3**

2 Introduction **3**

3 Model Creation **5**
3.1 Data 5
3.2 Model Training 6

4 Results **7**
4.1 Model Performance 8
4.2 Threshold Sensitivity Analysis 12
4.3 Model Efficiency and Deployment Performance 17
4.4 Explainability 20

5 Final Model Creation **21**
5.1 Corrected Dataset Creation 22
5.2 Large-Scale Model Study 23
5.3 Final Training Results and Model Selection 24
5.4 Held-Out Source Validation and Deployment Comparison 26
5.4.1 Per-Source Validation Results 26
5.4.2 Aggregate Validation Performance and Threshold Sensitivity 27
5.4.3 Qualitative Validation Example 27
5.4.4 CPU–GPU and End-to-End Pipeline Timing 30

6 Conclusion **31**
6.1 Future Directions and Next Steps 33

1 Preambule

This document provides an overview of the results obtained from the road extraction project. Its primary purpose is to present the technical findings of the developed model, including the rationale behind the selected architecture, model performance and evaluation results. A separate technical guide is also provided, offering detailed instructions on how to deploy and use the tool within ArcGIS.

2 Introduction

Roads, whether paved, gravel, or intended for logging, are a fundamental component of modern society. Road infrastructure has a profound impact on economic development and competitiveness by enabling economic growth. They facilitate the transport of goods and people across vast regions, supporting complex economic and social structures. Logging roads, consisting of dirt or gravel surfaces, are used by logging companies to access sites and transport materials. However, roads also profoundly affect local ecosystems by altering habitats, increasing animal mortality through collisions, and disrupting migration and mating patterns. Runoff from road salting and large-scale land clearing can further degrade surrounding flora and fauna. Compacted soil caused by machinery, such as that used during logging operations, has a negative impact both economically and environmentally. The negative impacts of these roads can be mitigated through improved management and monitoring, which remains a complex task due to the involvement of multiple organizations. On-site visits to these roads are useful for various purposes, such as analyzing their environmental and economic impacts. These visits are also necessary to confirm road usability and current condition in order to assess their overall impact. This requires detailed maps of road networks to enable effective tracking and monitoring, which is challenging due to the quantity and diversity of roads. While paved roads (e.g., highways) are generally well documented by government agencies, many unmapped or poorly maintained roads, particularly logging roads, remain untracked. Since these roads are often privately owned and distributed across different corporations and organizations, aggregating their locations into a centralized map is challenging. Remote sensing provides a promising approach for large-scale road surveying without the need to aggregate and verify data from multiple sources. Remote sensing techniques, such as satellite and drone imagery, enable coverage of large areas quickly and efficiently without requiring on-site visits. However, due to the vast extent of logging and secondary roads, manually analyzing and mapping roads from remote sensing data remains challenging. Manual tracing and extraction can be accurate but are highly time-consuming, as they require tedious annotation across thousands of images. New-Brunswick, covering nearly 73,000 km^2, presents additional challenges due to the abundance of logging roads and continuously changing environmental conditions. This project explores the use of automated techniques to extract roads from remote sensing data in New-Brunswick, Canada. Artificial Intelligence, particularly Deep Learning (DL), has already demonstrated strong

performance in similar tasks. DL-based segmentation enables the extraction of objects from images, such as roads, with a high degree of accuracy and reliability. This project was divided into three phases: an exploratory phase, a large-scale study of DL algorithms, and a final fine-tuning and integration phase. The exploratory phase consisted of initial model training to assess the feasibility of the approach and yielded promising results. The second phase involved a large-scale study of DL using various algorithms and techniques on a larger annotated dataset. The results were highly promising and were used to generate the final dataset for fine-tuning and integration. Previous technical reports provided an overview of the exploratory and large-scale phases, including key results and performance. However, the focus of this report is placed on the final model, its performance, and the fine-tuning process used for integration. The outline of this technical report is presented below.

Report Outline

1. **Model Creation**
   This section presents the dataset, preprocessing pipeline, model architecture, training procedure and evaluation methodology used to develop the road segmentation model.
2. **Model Performance**
   This section evaluates the segmentation performance of the final model using metrics such as Accuracy, Precision, Recall, F1-score, IoU and PPA. Best, average and worst-performing cases are also analyzed to better understand model behaviour.
3. **Threshold Sensitivity Analysis**
   This section investigates the impact of different probability thresholds on segmentation performance. Precision, Recall, F1-score, IoU, false positive rates and false negative rates are analyzed to determine an appropriate deployment threshold.
4. **Model Efficiency and Deployment Performance**
   This section evaluates the computational efficiency of the proposed solution, including ONNX inference speed, tile processing times, mosaicking performance and large-scale deployment estimates for CSRNO and New-Brunswick.
5. **Explainability**
   This section presents Grad-CAM explainability analyses used to visualize the image regions contributing most strongly to road predictions. These results provide insight into model decision-making and help validate that the model focuses on meaningful road structures.

3 Model Creation

This section presents the proposed model configuration, including the dataset, preprocessing pipeline and training parameters used for road segmentation.

3.1 Data

DL-based segmentation models require annotated datasets, often consisting of thousands of samples, in order to achieve strong generalization performance. Earlier experiments and related work demonstrated that DL-based segmentation can effectively extract logging roads from high-resolution satellite imagery. As such, CCNB-Innov annotated large areas of interest containing logging roads, paved roads and other transportation pathways. These samples were manually annotated and verified, increasing the reliability and validity of the dataset. The selected regions were collected from the CSRNO area and primarily consisted of agricultural, forested and rural environments.

In total, 39 regions were selected based on various criteria and manually annotated. Each case covered approximately 196 km^2, resulting in a total annotated area of approximately 7644 km^2. The imagery was obtained using the World Imagery layer within ArcGIS, which combines imagery from multiple satellite and aerial imagery databases. Using a cell size of 2, the imagery corresponded to a spatial resolution of approximately 2 meters per pixel and primarily included data from the Maxar WorldView-02 satellite. Figure 1 presents an example of the imagery used for annotation within ArcGIS.

Figure 1: Example of high-resolution imagery used for road annotation in ArcGIS.

Once annotated, the data were uploaded and combined within ArcGIS to generate corre-

sponding segmentation masks for each selected region. The imagery and masks were then exported using a cell size of 2 and an image size of 1024×1024 pixels. This corresponds to an approximate spatial coverage of 2.048 km $\times$ 2.048 km per image tile, enabling large-area coverage while maintaining relatively high spatial resolution. The export process consisted of extracting 1024×1024 image tiles using horizontal and vertical strides of 512 pixels, enabling overlap between neighbouring tiles and increasing data diversity. In total, 2,592 matching image and annotation pairs were exported. The dataset was subsequently divided using an 80:20 split for training and validation.

3.2 Model Training

Various segmentation architectures are available with different levels of complexity and numbers of parameters. Related work has demonstrated that transformer-based and larger vision architectures can sometimes provide slightly higher segmentation performance. However, these approaches often require substantially higher computational resources and significantly larger model sizes. In this application, pixel-level boundary precision is not the primary objective. Instead, the main goal is to efficiently detect and extract the maximum number of roads over large geographic regions. As such, model selection must balance segmentation performance, computational efficiency and deployment practicality.

Based on these considerations, a U-Net architecture with a ResNet101 encoder was selected. This architecture provided strong segmentation performance while maintaining a relatively lightweight and practical model size compared to more computationally demanding architectures. The selected model can also be trained or fine-tuned locally on consumer-grade hardware, enabling rapid experimentation and optimization without requiring specialized infrastructure. The model was trained using the AdamW optimizer for 30 epochs with a learning rate and weight decay of 1×10^{-4}. The input consisted of 1024×1024 RGB image patches normalized using ImageNet statistics. Input pixel values were first scaled between 0 and 1, followed by normalization using a mean of $[0.485, 0.456, 0.406]$ and a standard deviation of $[0.229, 0.224, 0.225]$.

This normalization strategy helped reduce variations caused by differences in imagery quality, illumination conditions and sensor characteristics while improving overall training stability across heterogeneous data sources. The selected image size also enabled the model to capture broader contextual information surrounding road networks while remaining feasible to train and deploy on consumer-grade GPUs. Several data augmentation techniques were implemented to improve model generalization and robustness. These augmentations included horizontal and vertical flipping, random rotations, affine transformations, brightness and contrast adjustments, Gaussian noise injection and motion blur augmentations.

4 Results

This section evaluates the performance of the proposed road segmentation model in terms of segmentation accuracy, robustness and deployment suitability. Several metrics were used to provide a comprehensive assessment of model performance, including Accuracy, Precision, Recall, F1-score, Intersection over Union (IoU), False Positive Rate (FPR), False Negative Rate (FNR) and Pixel Proximity Accuracy (PPA).

Accuracy measures the proportion of correctly classified pixels and is defined by Equation 1.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where TP, TN, FP and FN represent true positives, true negatives, false positives and false negatives, respectively.

Precision evaluates the reliability of road predictions and is calculated using Equation 2.

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

Recall, also referred to as Road Pixel Accuracy, measures the proportion of annotated road pixels successfully detected by the model and is defined by Equation 3.

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

The F1-score combines Precision and Recall into a single metric and is defined by Equation 4.

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \tag{4}$$

Intersection over Union (IoU) evaluates the overlap between predicted and annotated road pixels and is calculated using Equation 5.

$$IoU = \frac{TP}{TP + FP + FN} \tag{5}$$

The False Positive Rate (FPR) and False Negative Rate (FNR) quantify the proportion of incorrectly classified pixels and are defined by Equations 6 and 7, respectively.

$$FPR = \frac{FP}{FP + TN} \tag{6}$$

$$FNR = \frac{FN}{FN + TP} \tag{7}$$

In addition to conventional segmentation metrics, Pixel Proximity Accuracy (PPA) was used to evaluate prediction quality while accounting for small spatial localization errors. Road extraction is particularly sensitive to minor positional offsets because roads are narrow linear structures and even small shifts between annotations and predictions can significantly reduce strict pixel-level metrics. PPA addresses this limitation by introducing a tolerance buffer around annotated road pixels.

PPA$n0$ corresponds to the conventional precision metric, where a predicted road pixel is considered correct only if it exactly overlaps an annotated road pixel. For PPA$n1$, a prediction is considered correct if it falls within one pixel of an annotated road pixel. Similarly, PPA$n2$ and PPA$n5$ allow localization tolerances of two and five pixels, respectively. Consequently, PPA provides additional insight into whether prediction errors originate from complete road omission or from minor spatial misalignments.

The following subsections present the quantitative performance of the proposed model, followed by analyses of threshold sensitivity, computational efficiency and model explainability.

4.1 Model Performance

During training, the model achieved a best validation F1-score of 74.50%, with a validation IoU of 59.36%, Precision of 72.09%, Recall of 77.07% and an overall Accuracy of 99.16% after 30 epochs. The best-performing epoch was subsequently selected for further analyses and integrated into the developed GUI for operational testing and large-scale prediction. This model was trained prior to the correction of a small number of annotation inconsistencies related to paved roads being incorrectly labeled within the dataset. Nevertheless, the overall performance was largely unaffected, as these annotation errors represented only a small portion of the training data and were rarely encountered throughout the dataset. A final updated model will be provided in the future once the corrected annotations have been incorporated and the training process has been repeated. Although the expected impact is relatively minor, retraining with the corrected annotations should improve dataset consistency and overall model reliability.

Table 1: Average segmentation metrics across all test cases.

Metric	**Value (%)**
Accuracy	98.89
Precision	63.69
Recall (Road Pixel Accuracy)	66.67
F1-Score	64.98
IoU	48.51
PPA_{n0}	63.69
PPA_{n1}	68.66
PPA_{n2}	73.23
PPA_{n5}	82.46
False Positive Rate	0.60
False Negative Rate	33.33

Using all available cases, including both training and validation samples, the model achieved the average metrics presented in Table 1. Overall, the results demonstrate that the proposed model was capable of successfully extracting the majority of agricultural and forestry roads while maintaining a very low number of false detections The average Accuracy of 98.89% indicates that the model correctly classified nearly all pixels within the evaluated images. However, because road pixels represent only a small fraction of the total image area, Accuracy alone can provide an overly optimistic assessment of segmentation performance. Consequently, metrics such as Precision, Recall, F1-score and IoU provide a more representative evaluation of road extraction quality. The average F1-score of 64.98% and IoU of 48.51% indicate a moderate-to-strong overlap between predicted and annotated roads across all test cases. Similarly, the average Recall of 66.67% shows that approximately two-thirds of all annotated road pixels were successfully detected by the model. The average Precision of 63.69% indicates that most predicted road pixels corresponded to actual roads, while maintaining a relatively low number of false detections. This behaviour is further supported by the extremely low False Positive Rate of only 0.60%, demonstrating that the model rarely classifies background regions as roads. On the other hand, the relatively high FNR of 33.33% indicates that approximately one-third of annotated road pixels were classified as background. While this may initially appear significant, it should be interpreted alongside the PPA metrics. Road extraction is particularly sensitive to small spatial offsets because roads occupy only a few pixels in width. Consequently, predictions that are visually correct may still be penalized as false negatives if they are shifted by only a small number of pixels relative to the annotations. Additionally, pixel-perfect annotations were not required during dataset creation, introducing a degree of uncertainty along road boundaries. The Pixel Proximity Accuracy (PPA) metrics provide additional insight into segmentation quality by accounting for small localization errors. While the strict pixel-level score PPA_{n0} achieved 63.69%, the value increased to 68.66%, 73.23% and 82.46% for tolerances of 1, 2

and 5 pixels, respectively. This substantial increase suggests that many apparent segmentation errors originate from small spatial offsets rather than complete road omission. The substantial increase from 63.69% PPA_{n0} to 82.46% PPA_{n5} also suggests that many of the false negatives are attributable to minor localization differences rather than complete failures to detect roads.

Such behaviour is expected in road segmentation tasks because roads represent narrow linear structures where even small prediction shifts can significantly affect conventional pixel-level metrics. Furthermore, pixel-perfect annotations were not required during dataset creation, as the primary objective was to identify road locations rather than precisely delineate every road boundary. Consequently, some disagreement between predictions and annotations may originate from annotation variability rather than true segmentation errors. The strong increase observed between PPA_{n0} and PPA_{n5} therefore indicates that the model generally identifies the correct road locations even when exact pixel alignment is not achieved. Another important observation is the relationship between Recall and the False Negative Rate. While the model successfully detected approximately 66.67% of annotated road pixels, approximately 33.33% remained undetected. These missed detections generally correspond to narrow roads, visually ambiguous roads, partially occluded roads or road segments with limited contrast relative to the surrounding environment. Nevertheless, the low False Positive Rate indicates that the model remains relatively conservative, favouring prediction reliability over excessive road generation.

Overall, the results indicate that the proposed model achieves a strong balance between detection capability and prediction reliability. The model successfully extracts most forestry and agricultural roads while maintaining a low level of false detections, making it suitable for large-scale operational road mapping applications. The PPA results further suggest that a considerable portion of the measured errors are attributable to minor localization differences rather than complete failures to identify roads. To further analyze model behaviour, the best-performing case, an average-performing case and the worst-performing case were selected based on their F1-score values. The best-performing sample was `case 93`, while the lowest-performing sample was `case 125`. A representative average-performing sample, `case 96`, was also selected since its metrics closely matched the global averages presented in Table 1. The quantitative results for these three cases are presented in Table 2.

Table 2: Comparison between selected test cases based on F1-score.

Metric	**Best** (`case 93`)	**Avg.** (`case 96`)	**Worst** (`case 125`)
Accuracy	99.35	98.89	98.49
Precision	71.87	63.69	49.98
Recall	80.71	66.67	41.97
F1-Score	76.03	64.98	45.62
IoU	61.33	48.51	29.55

Figures 2a, 2b and 2c respectively present Case 93, Case 96 and Case 125. In these visualizations, true positive road pixels are represented in green, false negatives in red and false positives in orange, while correctly identified background pixels are not shown.

(a) Best-performing case (`case 93`).

(b) Average-performing case (`case 96`).

(c) Worst-performing case (`case 125`).

Figure 2: Examples of road segmentation performance for the best, average and worst-performing cases. True positive road pixels are represented in green, false negatives in red and false positives in orange.

Case 93 demonstrates the maximum performance achieved by the model. Most roads were correctly extracted with very limited false positive and false negative predictions. This behaviour is reflected by the F1-score of 76.03% and IoU of 61.33%. The model successfully identified the majority of agricultural and forestry roads while maintaining strong continuity along road segments and preserving thin linear structures. Only a few isolated prediction errors can be observed, generally occurring near road intersections, narrow road extremities or visually ambiguous regions.

Case 96 represents a typical or average-performing example. The metrics of this case closely match the global averages presented in Table 1, making it representative of the expected operational behaviour of the model. Most roads were successfully segmented with only moderate false positive and false negative predictions. The model was generally capable of following road network geometry while preserving road continuity and localization accuracy. Most prediction errors occurred along road boundaries or within short disconnected segments rather than through complete road omission. This behaviour suggests that the model can reliably extract the majority of forestry and agricultural roads even under moderately challenging conditions.

Case 125 represents the lowest-performing sample among the evaluated cases. In this example, a larger amount of false negatives and false positives can be observed, resulting in lower Precision, Recall, F1-score and IoU values. The primary source of error appears to be related to major paved roads and highways rather than forestry or agricultural roads. Visual inspection suggests that these errors are strongly influenced by annotation inconsis-

tencies, as paved roads were not consistently annotated as target roads within the dataset. Consequently, several correctly predicted paved roads were penalized as false positives, while unannotated paved road segments contributed to false negatives. Despite these inconsistencies, the model still performed relatively well on the intended forestry and agricultural road classes. This observation suggests that a portion of the measured errors likely originates from annotation quality and class-definition inconsistencies rather than purely from segmentation limitations.

Overall, the qualitative and quantitative analyses demonstrate that the proposed model consistently extracts forestry and agricultural roads across a wide range of environmental conditions. While performance naturally varies between cases, the model maintains relatively stable behaviour and produces useful road predictions suitable for large-scale operational deployment.

4.2 Threshold Sensitivity Analysis

To further analyze the behaviour of the proposed road segmentation model, a threshold sensitivity analysis was conducted using Case 96. The threshold is the value which a model will determine if a pixel is road or background, ranging from 0 to 1. Most often, this value is set at 0.5 and values above are positive, while below are negative. The objective of this experiment was to evaluate the impact of the prediction threshold on the balance between false positives and false negatives while also analyzing the evolution of common segmentation metrics such as Precision, Recall, F1-score and IoU. The experiment was performed using the final ONNX deployment model integrated within the ArcGIS Pro pipeline. Predictions were generated once using the probability output of the model and evaluated using thresholds ranging from 0.00 to 1.00 with increments of 0.05. The ground-truth raster was aligned to the prediction raster resolution and extent prior to evaluation to ensure correct pixel-level comparisons.

The results demonstrated that threshold selection has a relatively limited impact on overall segmentation performance within a broad intermediate range. Figure 3 shows the evolution of Precision, Recall, F1-score and IoU across all evaluated thresholds. At very low thresholds, particularly 0.00, nearly all pixels are classified as roads, resulting in extremely high Recall but very poor Precision and IoU. Conversely, at a threshold of 1.00, almost no road pixels are detected, causing Recall, F1-score and IoU to collapse toward zero. However, between approximately 0.10 and 0.90, segmentation performance remains remarkably stable. The best F1-score and IoU were obtained at a threshold of 0.40. At this threshold, the model achieved an F1-score of 69.87%, an IoU of 53.69%, a Precision of 67.88% and a Recall of 71.98%. The default threshold of 0.50 produced nearly identical performance, with an F1-score of 69.84% and an IoU of 53.66%. This negligible difference indicates that the proposed model is relatively robust to moderate threshold variations and does not require precise threshold tuning to maintain stable segmentation performance.

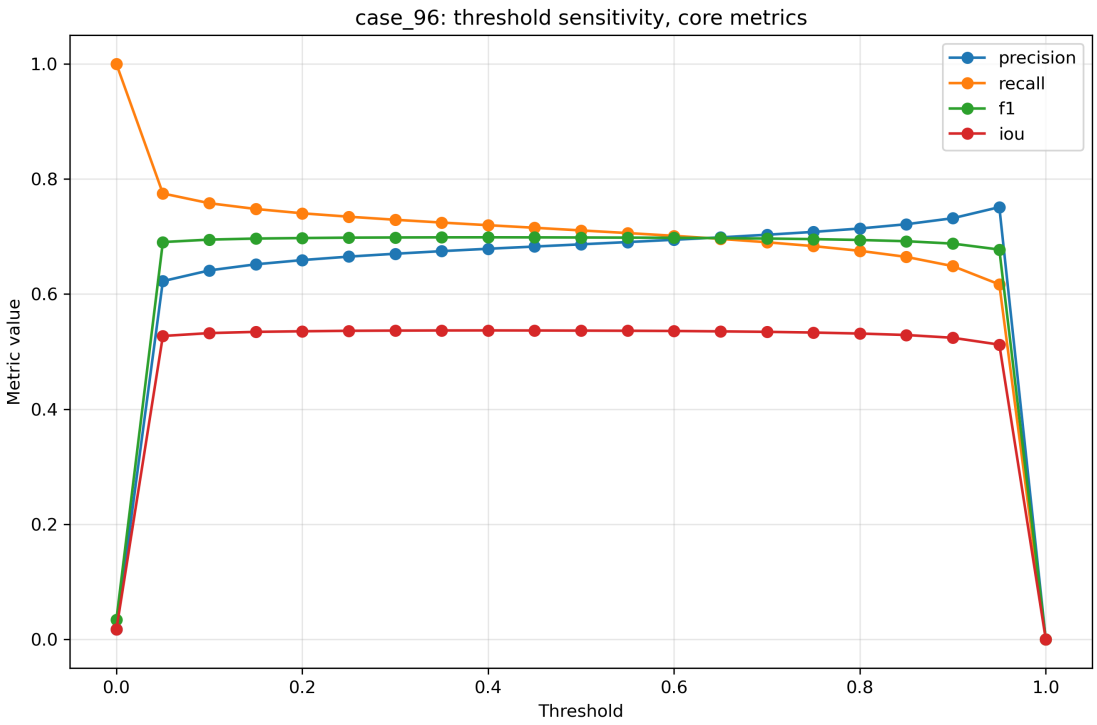


Figure 3: Evolution of Precision, Recall, F1-score and IoU for Case 96 across different threshold values.

The relationship between the False Positive Rate (FPR) and False Negative Rate (FNR) is illustrated in Figure 4. As expected, increasing the threshold gradually reduces the FPR while simultaneously increasing the FNR. At low thresholds, the model tends to over-segment roads, producing additional false positive detections. At higher thresholds, the model becomes more conservative, reducing false positives but missing more road segments. The FPR remained very low throughout most thresholds, generally below 1%, while the FNR increased progressively as the threshold increased. This behaviour is expected, as higher thresholds make the model more conservative, reducing false detections at the cost of missing additional road pixels.

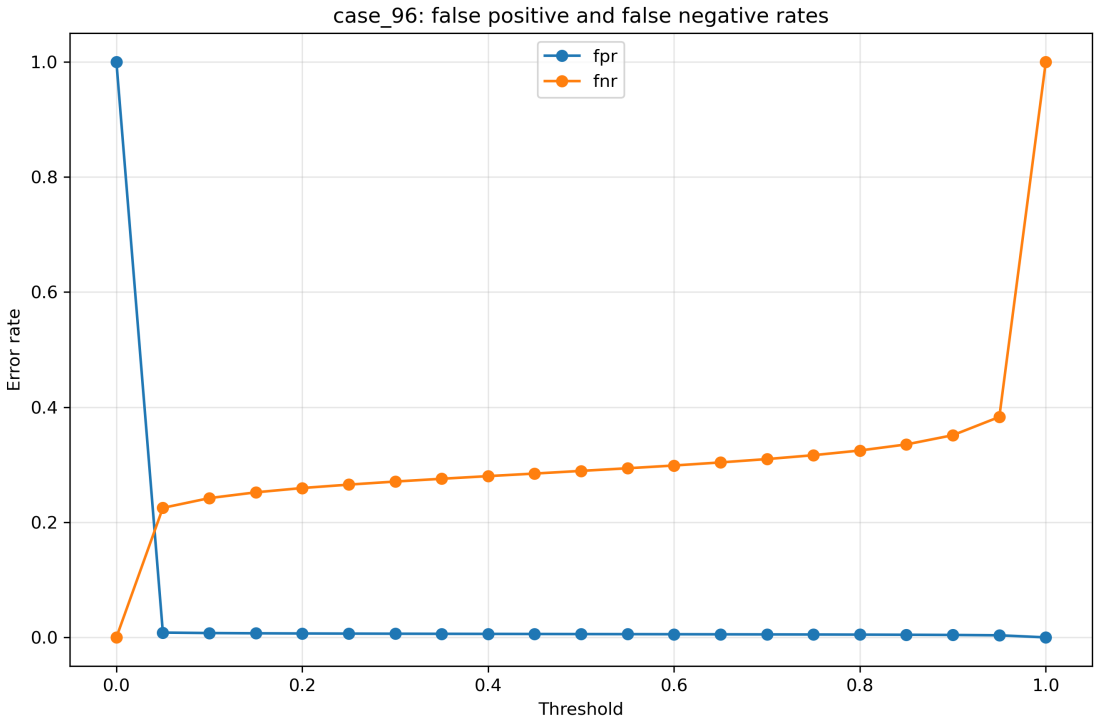


Figure 4: False Positive Rate and False Negative Rate for Case 96 across different threshold values.

Table 3 summarizes the quantitative results obtained for each evaluated threshold. The table further confirms that the model maintains relatively stable performance across a broad threshold range. The highest F1-score and IoU values were obtained at thresholds between approximately 0.35 and 0.50, while extreme threshold values significantly degraded segmentation performance. Very low thresholds resulted in excessive false positives and over-segmentation, whereas high thresholds progressively increased false negatives and under-segmentation.

Table 3: Threshold sensitivity results for Case 96.

Threshold	Precision (%)	Recall (%)	F1-score (%)	IoU (%)	FPR (%)	FNR (%)	Accuracy (%)
0.00	1.70	100.00	3.40	1.70	100.00	0.00	1.70
0.05	62.30	77.60	69.10	52.80	0.90	22.40	98.70
0.10	64.00	75.90	69.50	53.30	0.80	24.10	98.80
0.15	65.10	75.10	69.70	53.50	0.70	24.90	98.80
0.20	65.70	74.30	69.70	53.50	0.70	25.70	98.80
0.25	66.20	73.60	69.70	53.50	0.70	26.40	98.80
0.30	66.70	73.00	69.70	53.60	0.60	27.00	98.90
0.35	67.30	72.40	69.80	53.60	0.60	27.60	98.90
0.40	67.90	72.00	69.87	53.69	0.60	28.00	98.90
0.45	68.30	71.50	69.90	53.70	0.60	28.50	98.90
0.50	68.70	71.10	69.84	53.66	0.60	28.90	98.90
0.55	69.10	70.50	69.80	53.60	0.60	29.50	98.90
0.65	70.00	69.60	69.70	53.50	0.50	30.40	98.90
0.70	70.40	69.10	69.70	53.50	0.50	30.90	98.90
0.75	70.80	68.50	69.60	53.40	0.50	31.50	98.90
0.80	71.30	67.80	69.30	53.20	0.40	32.20	98.90
0.85	71.90	66.80	69.20	52.90	0.40	33.20	98.90
0.90	73.10	65.10	68.80	52.50	0.40	34.90	98.90
0.95	75.00	61.80	67.80	51.30	0.30	38.20	98.90
1.00	0.00	0.00	0.00	0.00	0.00	100.00	98.30

A visual confusion analysis for the best-performing threshold (0.40) is presented in Figure 5. True positives are shown in green, true negatives in black, false positives in orange and false negatives in red. The visualization demonstrates that the majority of road segments are correctly identified, while most errors occur along road boundaries, narrow road sections and partially ambiguous regions. False positives are generally sparse and localized, whereas false negatives mainly occur on weaker, visually unclear or partially occluded road segments. Most of the observed errors are therefore associated with localization and boundary delineation rather than complete failures to identify roads. Overall, the confusion visualization confirms the quantitative results and demonstrates the ability of the proposed model to maintain consistent segmentation quality throughout the evaluated region.

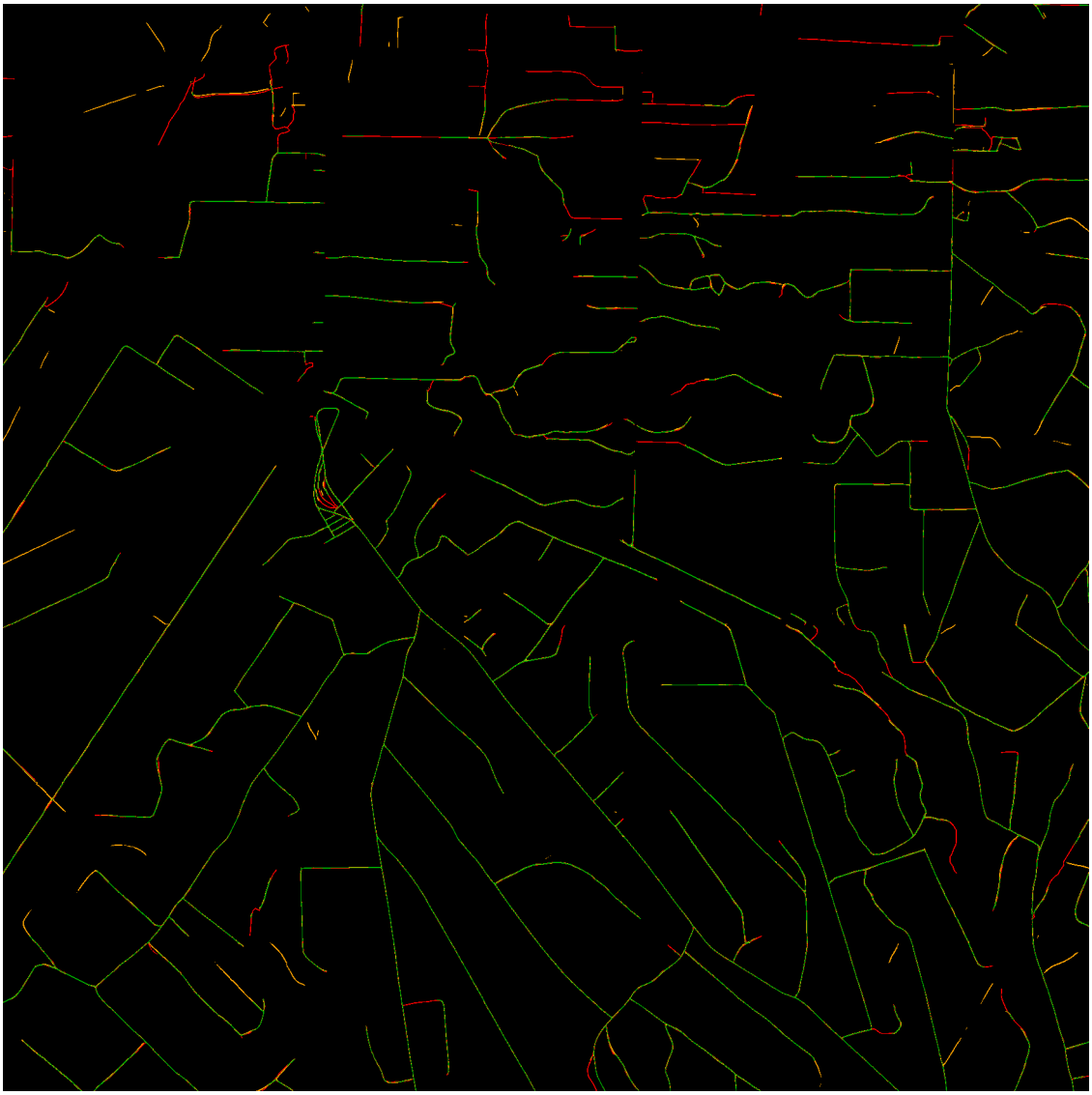


Figure 5: Confusion visualization for Case 96 at the best-performing threshold (0.40). Green corresponds to true positives, black to true negatives, orange to false positives and red to false negatives.

Overall, the threshold sensitivity analysis demonstrates that the proposed road segmentation model remains highly stable across a broad threshold range. The F1-score remains close to 70% between approximately 0.10 and 0.90, while the IoU remains near 53% throughout the same interval. Although the highest performance was obtained near a threshold of 0.40, the difference compared to the default threshold of 0.50 is negligible. Consequently, a threshold value of 0.50 was retained for deployment due to its balanced Precision–Recall trade-off, stable performance characteristics and compatibility with common segmentation practices.

Figure 6 presents the original RGB imagery and the corresponding probability distribution generated by the proposed segmentation model for Case 96. The left image shows the RGB

input provided to the model, while the right image presents the continuous probability map produced prior to thresholding. Rather than directly generating a binary road mask, the model first assigns a confidence value to each pixel representing the likelihood that the corresponding location belongs to a road. Dark blue regions correspond to probabilities close to 0, indicating strong confidence that the pixel belongs to the background class, while bright yellow regions correspond to probabilities close to 1, indicating strong confidence that the pixel belongs to a road. Intermediate purple and orange regions represent uncertain predictions and are generally observed near road boundaries, intersections or visually ambiguous regions. The probability map highlights the locations that the model considers most likely to correspond to roads and demonstrates that most road segments are assigned relatively high confidence values while most background regions remain close to zero. This behaviour explains why moderate threshold adjustments have only a limited impact on segmentation performance, as the majority of pixels are confidently classified and remain unchanged across a broad range of threshold values. The main issue in this sample is the complexity of the road network and overlapping colors between road and background pixels.

(a) RGB input image.

(b) Model probability distribution.

Figure 6: RGB imagery and corresponding probability distribution generated by the model for Case 96.

4.3 Model Efficiency and Deployment Performance

Model efficiency is an important consideration, particularly during deployment and operational integration, and heavily influenced the final model selection. Although larger architectures such as SegFormer and other transformer-based models could potentially achieve slightly higher segmentation performance, these approaches are substantially larger and require significantly more computational resources. Such models generally involve higher mem-

ory usage, slower inference times and more complex deployment pipelines, making them less suitable for rapid large-scale prediction and practical integration into lightweight GIS-based tools. As such, a U-Net architecture with a ResNet101 encoder was selected due to its relatively moderate size, stable performance and ease of implementation for further development and deployment. This architecture provides a balance between segmentation accuracy and computational efficiency while remaining compatible with consumer-grade hardware. The selected model contains approximately 59 million trainable parameters and over one hundred convolutional layers through the encoder-decoder structure, making it reasonably sized compared to transformer-based alternatives while still maintaining strong segmentation capabilities.

Performance experiments were conducted on consumer-grade hardware equipped with an NVIDIA RTX 3060 GPU, 32 GB of RAM and an Intel i9-10900 processor, representing a moderate desktop computer configuration accessible to many organizations and researchers. To evaluate deployment efficiency, Case 102 was used as a benchmark scenario. This case consisted of 49 image tiles covering an area of approximately 196 km^2. The complete deployment pipeline included four primary stages:

- tile export from ArcGIS
- ONNX prediction
- raster mosaicking
- clipping and post-processing

Using batch inference with a batch size of 8 and CUDA acceleration, the complete processing pipeline required only 48.86 seconds to process the full 196 km^2 study area. This corresponds to an average processing speed of approximately 1.00 second per tile and 0.25 seconds per km^2. The tile export stage required 13.40 seconds, representing approximately 27% of the total processing time. The ONNX prediction stage required 18.21 seconds, accounting for approximately 37% of the complete pipeline. However, a more detailed analysis reveals that only 5.86 seconds were spent performing actual neural network inference. The remaining prediction time consisted primarily of preprocessing operations (2.31 seconds) and prediction export and saving operations (9.74 seconds). This observation highlights an important characteristic of the deployment pipeline: the deep learning model itself is not the primary computational bottleneck. Pure ONNX inference represented only approximately 12% of the total pipeline execution time and achieved an average inference speed of 0.12 seconds per tile. In comparison, preprocessing, file generation and raster export operations collectively required more time than the neural network inference itself. The mosaicking stage required 5.50 seconds, while the final clipping and cleaning operations required 10.36 seconds. Together, these GIS processing operations accounted for approximately one-third of the total execution time. Consequently, future deployment optimizations are likely to benefit more from improvements to data handling, raster I/O and ArcGIS processing workflows than from additional model-level optimizations.

Table 4: Deployment efficiency results for Case 102 using ONNX inference and CUDA acceleration.

Metric	**Value**
Hardware	RTX 3060 / i9-10900 / 32 GB RAM
Case Area	196 km^2
Number of Tiles	49
Tile Size	(1024 × 1024)
Batch Size	8
Processing Times	
Export Time	13.40 s
Prediction Time	18.21 s
Preprocessing	2.31 s
ONNX Inference	5.86 s
Post-processing + Save	9.74 s
Mosaic Time	5.50 s
Clip + Clean Time	10.36 s
Total Pipeline Time	**48.86 s**
Average Processing Times	
Export per Tile	0.27 s
Prediction per Tile	0.37 s
Preprocessing per Tile	0.05 s
ONNX Inference per Tile	0.12 s
Post-processing per Tile	0.20 s
Mosaic per Tile	0.11 s
Total per Tile	**1.00 s**
Total per km^2	0.25 s
Estimated Processing Times	
CSRNO Coverage	(∼30) min
New-Brunswick Coverage	(∼5) h

Table 4 summarizes the deployment efficiency results obtained for Case 102. These results demonstrate that the proposed approach can efficiently process large geographic regions while maintaining practical deployment speeds on moderate consumer-grade hardware. The relatively low computational requirements also enable future integration into operational GIS workflows and potential near-real-time applications. By extrapolating these timing results, the proposed system could process the entire CSRNO region, which has an approximate area of $7,196.83$ km^2, in roughly 30 minutes under similar hardware and deployment conditions. Similarly, extrapolation to the entire province of New-Brunswick, with an approximate area of $72,908$ km^2, suggests that complete province-wide prediction could theoretically be achieved in approximately 5 hours using a single consumer-grade GPU system. Although these estimates provide a useful indication of the scalability of the proposed approach, they

are based solely on the total area covered and assume ideal processing conditions. In practice, actual processing times would likely be higher due to factors such as tile overlap, buffering operations, irregular region boundaries, image export overhead and additional GIS processing requirements. Furthermore, some deployments may require smaller strides or overlapping tiles to reduce edge artifacts and improve prediction continuity, increasing the total number of processed tiles. Consequently, a more realistic estimate would be approximately 1 hour for complete CSRNO coverage and approximately 10 hours for province-wide processing across New-Brunswick using the current deployment workflow and hardware configuration. Even under these more conservative estimates, the proposed approach remains highly practical for large-scale road extraction and environmental monitoring applications.

These results highlight the practical scalability of the proposed solution and demonstrate that large-scale forestry and agricultural road extraction can be performed efficiently without requiring high-end server infrastructure, cloud computing resources or specialized computational clusters. More heavily optimized implementations, particularly those avoiding GUI-based operations and relying exclusively on direct ONNX inference pipelines, could further improve deployment efficiency. Likewise, stronger hardware configurations equipped with more powerful GPUs, faster storage devices and increased memory bandwidth would further reduce processing times. Currently, one of the primary bottlenecks remains World Imagery extraction within ArcGIS. Unlike the neural network inference stage, which executes locally on the GPU, imagery must first be requested from remote ArcGIS servers during tile export operations. Consequently, network latency, server response times and raster export operations contribute substantially to the overall processing time. In practical deployments, storing imagery locally or processing pre-downloaded datasets would likely provide some of the largest performance improvements within the entire workflow.

4.4 Explainability

To provide additional insight into model behaviour, explainability analyses were conducted. As DL models are generally considered black-box approaches, explainability techniques can help better understand how convolutional neural networks generate predictions and which image regions contribute the most to the final output. Grad-CAM was used to generate explainability visualizations. This approach allows the selection of a convolutional layer within the network, typically one of the final encoder layers, to visualize the most activated regions associated with the prediction. By backpropagating gradients through the selected layer, Grad-CAM highlights the image pixels and structures that contributed the most to the final segmentation output. Explainability methods have shown strong benefits across multiple domains, particularly in applications where model interpretability and validation are important. In remote sensing applications, these techniques can help validate whether the model focuses on meaningful structures such as roads, edges and textures rather than unrelated background information. Figure 7 presents an example including the RGB image, the predicted segmentation mask and the corresponding Grad-CAM explainability visualization.

The results show that the model primarily focuses on linear road structures and surrounding road boundaries when generating predictions. Large portions of the road network are identified as highly important regions, while surrounding vegetation and background areas generally receive lower activation values. The visualizations also suggest that earlier extracted road-like structures are progressively refined through deeper convolutional layers to generate the final segmentation output. These observations indicate that the model learned meaningful spatial and contextual representations associated with forestry and agricultural roads.

[H]

(a) Original image (Case 158) **(b)** Predicted mask **(c)** Explainability overlay

Figure 7: Example of road segmentation predictions and Grad-CAM explainability visualizations for three randomly selected samples.

5 Final Model Creation

The first phase of the project, presented in the preceding sections, established the initial road extraction workflow, evaluated the first U-Net model and identified several annotation inconsistencies through quantitative testing and explainability analysis. A second phase of the work was therefore conducted to construct a final model using a revised and more consistent dataset. This phase was divided into two main components: first, the creation of a corrected road dataset from 59 large source tiles; and second, a broader model-selection study in which ten architecture and loss-function configurations were trained and compared under the same validation protocol.

Importantly, the models described in this section were trained as new road-segmentation experiments and did not resume from the previously selected road model. The encoder weights were initialized from ImageNet-pretrained weights, while the road-segmentation decoder and optimization process were trained using the newly generated dataset.

5.1 Corrected Dataset Creation

A revised dataset containing 59 large image–annotation pairs was prepared for the final training phase. The road annotations associated with these large source tiles were reviewed and corrected by CCNB Innov before model training. This correction step was intended to reduce the annotation inconsistencies observed during the first phase of the project and to provide a more reliable reference dataset for final model selection.

To avoid spatial leakage between training and validation, the split was performed at the level of the original large source tiles *before* any cropping operation. Using a 90:10 source-level split, 53 of the 59 large tiles were assigned to training and 6 were reserved for validation. Consequently, no crop extracted from a given large source tile could appear in both subsets.

The final training dataset was generated using road-aware sampling. Each training source contributed 400 crops of 1024×1024 pixels, resulting in a total of 21,200 training image–mask pairs. Rather than using a dense overlapping grid dominated by background, the sampling procedure attempted to make approximately 93% of the training crops road-containing. A crop was considered road-targeted when it contained at least 300 road pixels. The remaining crop positions were sampled without restriction; they could contain no road pixels, a small amount of road or a large amount of road. This ensured that the dataset remained strongly focused on the target class without imposing an artificial requirement for empty background images.

The validation set was generated differently in order to preserve a representative spatial distribution. The six held-out source tiles were exhaustively cropped using 1024×1024 windows with a stride of 512 pixels, corresponding to 50% overlap. No road-aware filtering was applied to validation data. This produced 1,014 validation image–mask pairs. The resulting dataset therefore contained 21,200 training crops and 1,014 validation crops, while maintaining strict separation between the original source regions.

No augmented images were saved during dataset generation. Instead, augmentation was performed dynamically during training so that a different transformed version of the data could be presented at each epoch. The training pipeline used horizontal and vertical flips, random 90° rotations, affine transformations with rotations up to approximately $\pm 15°$, modest scaling and translation, brightness and contrast changes, gamma adjustments, small hue/saturation/value variations, Gaussian noise and occasional mild Gaussian blur. All geometric transformations were applied jointly to the imagery and road masks to preserve pixel-level alignment.

Table 5: Summary of the corrected final training dataset.

Dataset characteristic	Value
Original corrected source tiles	59
Training source tiles	53
Validation source tiles	6
Source-level split	90:10
Training crop size	1024×1024 pixels
Training crops per source	400
Total training crops	21,200
Road-target sampling objective	93%
Minimum road pixels for targeted crop	300
Validation crop size	1024×1024 pixels
Validation stride	512 pixels
Total validation crops	1,014
Pre-generated augmentation	None
Training augmentation	On-the-fly

5.2 Large-Scale Model Study

Following creation of the corrected dataset, a larger model-selection study was performed. Five segmentation architectures were investigated: U-Net, U-Net++, DeepLabV3+, MAnet and SegFormer. The convolutional architectures used a ResNet101 encoder, while SegFormer used the MiT-B5 encoder. Three loss formulations were considered across the experiments: Dice+BCE, Dice+Focal and Tversky+Focal. The Tversky formulation used $\alpha = 0.30$ for false positives and $\beta = 0.70$ for false negatives, thereby placing greater emphasis on missed road pixels.

Table 6 summarizes the ten model configurations included in the study.

Table 6: Architecture and loss-function configurations evaluated during final model selection.

Run	Architecture	Loss
01	U-Net + ResNet101	Dice + BCE
02	U-Net + ResNet101	Dice + Focal
03	U-Net++ + ResNet101	Dice + Focal
04	U-Net++ + ResNet101	Tversky + Focal
05	DeepLabV3+ + ResNet101	Dice + Focal
06	DeepLabV3+ + ResNet101	Tversky + Focal
07	MAnet + ResNet101	Dice + Focal
08	MAnet + ResNet101	Tversky + Focal
09	SegFormer + MiT-B5	Dice + Focal
10	SegFormer + MiT-B5	Tversky + Focal

All models were trained using 1024×1024 inputs and ImageNet normalization. Optimization used AdamW with separate learning rates for encoder and non-encoder parameters. The initial encoder learning rate was 2×10^{-5}, while the remaining model parameters used 1×10^{-4}. A cosine annealing learning-rate scheduler was used with a minimum learning rate of 10^{-6}, weight decay of 10^{-4}, gradient clipping at 1.0 and a maximum of 50 epochs. Early stopping was applied when the validation F1-score at a threshold of 0.50 failed to improve for 12 consecutive epochs.

For the ResNet101-based models, the physical batch size was 2 with two gradient-accumulation steps, producing an effective batch size of approximately 4. For the two SegFormer MiT-B5 experiments, the physical batch size was reduced to 1 because of the larger memory requirement, while four accumulation steps maintained an effective batch size of approximately 4.

During training, the best checkpoint for each model was selected according to validation F1-score at a threshold of 0.50. After training, the best checkpoint was reloaded and evaluated at thresholds of 0.10, 0.25, 0.50, 0.75 and 0.90. The threshold producing the highest validation F1-score was then recorded for each experiment. This procedure allows model architecture and loss-function performance to be compared while also accounting for differences in probability calibration between models.

5.3 Final Training Results and Model Selection

Table 7 presents the validation results obtained from the ten-model study. The reported precision, recall, F1-score and IoU correspond to the best tested threshold for each successfully trained model.

Table 7: Validation results from the final ten-model training study.

Run	Architecture	Loss	Best epoch	Threshold	Precision	Recall	F1-score	IoU
01	U-Net + ResNet101	Dice + BCE	18	0.25	73.47%	76.75%	75.07%	60.09%
02	U-Net + ResNet101	Dice + Focal	3	0.50	74.47%	75.95%	75.20%	60.26%
03	U-Net++ + ResNet101	Dice + Focal	2	0.75	75.06%	75.02%	75.04%	60.05%
04	U-Net++ + ResNet101	Tversky + Focal	16	0.90	72.40%	78.22%	75.20%	60.26%
05	DeepLabV3+ + ResNet101	Dice + Focal	12	0.25	73.34%	76.05%	74.67%	59.58%
06	DeepLabV3+ + ResNet101	Tversky + Focal	2	0.75	67.69%	70.50%	69.06%	52.75%
07	MAnet + ResNet101	Dice + Focal	–	–	–	–	–	–
08	MAnet + ResNet101	Tversky + Focal	1	0.90	72.27%	74.73%	73.48%	58.08%
09	**SegFormer + MiT-B5**	**Dice + Focal**	**7**	**0.50**	**75.47%**	**76.43%**	**75.95%**	**61.22%**
10	SegFormer + MiT-B5	Tversky + Focal	8	0.75	72.51%	78.71%	75.48%	60.62%

Run 07, corresponding to MAnet with Dice+Focal loss, became numerically unstable during training and produced NaN losses. Its resulting zero-valued validation metrics were therefore not interpreted as meaningful segmentation performance and the run was excluded from model ranking.

Among the successfully trained configurations, SegFormer with the MiT-B5 encoder and Dice+Focal loss (Run 09) achieved the highest validation F1-score. Its best checkpoint was obtained at epoch 7, and the subsequent threshold sweep identified 0.50 as the best tested threshold. At this operating point, the model achieved a precision of 75.47%, recall of 76.43%, F1-score of 75.95% and IoU of 61.22%.

The comparison also shows that several ResNet101-based architectures reached similar F1-scores near 75%, indicating that the corrected dataset itself contributed to stable performance across multiple segmentation approaches. However, SegFormer produced the highest overall F1-score and IoU while maintaining a balanced precision–recall profile. The Tversky+Focal SegFormer variant achieved higher recall (78.71%) but lower precision (72.51%), resulting in a slightly lower F1-score of 75.48%. For this reason, the Dice+Focal SegFormer MiT-B5 configuration was selected as the final model for the second phase of the project.

The results also show that most models converged relatively quickly, indicating that the optimization process was generally stable and that the road-segmentation patterns present in the dataset were readily learnable by the tested architectures. Rapid convergence should not be interpreted as evidence that a global optimum was reached, but rather that additional training produced only limited improvements in validation performance. This behaviour was particularly evident for Runs 02, 03, 06 and 08, which reached their best validation performance within the first three epochs while still achieving results comparable to models that trained for considerably longer. The similarity in validation performance across architectures and loss functions suggests that, for this dataset, increasing model complexity alone provides only modest gains. Instead, model selection becomes a trade-off between segmentation accuracy, computational requirements, inference speed, and the intended deployment environment. Consequently, further performance improvements are likely to depend not only on architecture or loss-function tuning, but also on factors such as data quality, spatial

resolution, annotation consistency, sampling strategy, and inference configuration.

The results presented in this subsection are validation results from the held-out source-level validation set and should not be interpreted as independent external test performance. Independent case-based testing, threshold analysis and deployment benchmarking of the selected SegFormer model are evaluated separately.

5.4 Held-Out Source Validation and Deployment Comparison

After selection of the final SegFormer MiT-B5 model (run 09), an additional deployment-style validation was performed using the six original large source rasters that had been reserved for validation during corrected-dataset creation. The evaluation therefore used source regions that were not included among the 53 training source rasters. These results remain validation results, rather than an independent external test set, because the same six held-out source regions were available during model development and model selection.

The six large validation image–mask pairs were processed directly from the previously exported imagery rather than being re-exported as small ArcGIS chips. Each 7000×7000-pixel image was divided into 1024×1024-pixel windows using a stride of 512 pixels in both directions, corresponding to 50% overlap. This produced 169 inference windows per large source raster. For pixels covered by multiple windows, the predicted probabilities were accumulated and averaged before thresholding. The same procedure was applied to the final SegFormer MiT-B5 model and the previous U-Net ResNet101 model, providing a direct comparison under an identical inference pipeline. Ground-truth masks were interpreted as binary using the rule that every pixel value greater than zero represented road.

5.4.1 Per-Source Validation Results

Table 8 reports the best F1-Score operating point obtained for each model on each of the six large validation sources. The threshold sweep was performed from 0.00 to 1.00 in increments of 0.05. The final row gives the mean of the six case-specific best operating points.

Table 8: Per-source validation comparison between the final SegFormer MiT-B5 model and the previous U-Net ResNet101 model using 50% overlapping inference windows and probability averaging.

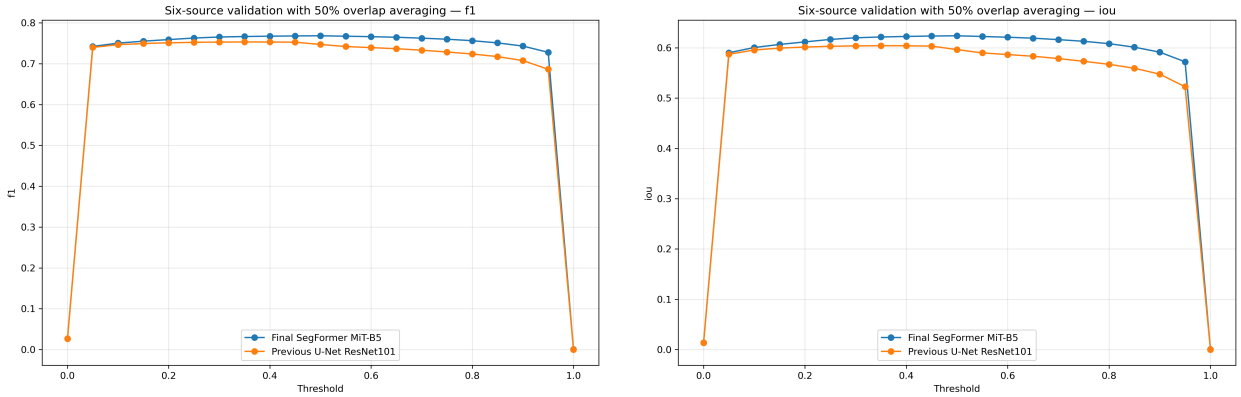
Validation source	SegFormer threshold	SegFormer F1	SegFormer IoU	U-Net threshold	U-Net F1	U-Net IoU
NB_Cellule_1183_000000000000	0.45	79.01%	65.30%	0.40	77.72%	63.56%
NB_Cellule_1336_000000000000	0.60	80.30%	67.08%	0.25	79.63%	66.16%
NB_Cellule_1589_000000000000	0.60	65.37%	48.56%	0.50	59.61%	42.46%
NB_Cellule_1647_000000000000	0.80	79.06%	65.37%	0.50	76.68%	62.17%
Qc_Case_743_000000000000	0.25	76.95%	62.53%	0.25	75.69%	60.88%
Qc_Cellule_455_000000000000	0.45	76.05%	61.35%	0.20	75.10%	60.13%
Mean	0.525	**76.12%**	**61.70%**	0.35	74.07%	59.23%

Across the six case-specific best operating points, SegFormer achieved a mean F1-score of 76.12% and mean IoU of 61.70%, compared with 74.07% F1-Score and 59.23% IoU for the

previous U-Net. SegFormer produced the higher case-specific best F1-score and IoU on each of the six validation sources. The lowest F1-Score values for both models occurred on `NB_Cellule_1589`, while the highest case-level F1-Score values occurred on `NB_Cellule_1336`.

5.4.2 Aggregate Validation Performance and Threshold Sensitivity

To obtain a single validation result across the complete held-out set, the confusion-matrix counts from all six large rasters were summed at each tested threshold before recalculating the metrics. This pixel-weighted aggregation represents a total of 294,000,000 evaluated pixels. The final SegFormer reached its highest aggregate F1-score at a threshold of 0.50, with precision 76.95%, recall 76.72%, F1-score 76.83% and IoU 62.38%. The previous U-Net reached its highest aggregate F1-score at a threshold of 0.35, with precision 74.82%, recall 75.82%, F1-score 75.32% and IoU 60.41%. The aggregate curves in Figure 8 show the corresponding F1-score and IoU values throughout the threshold sweep.



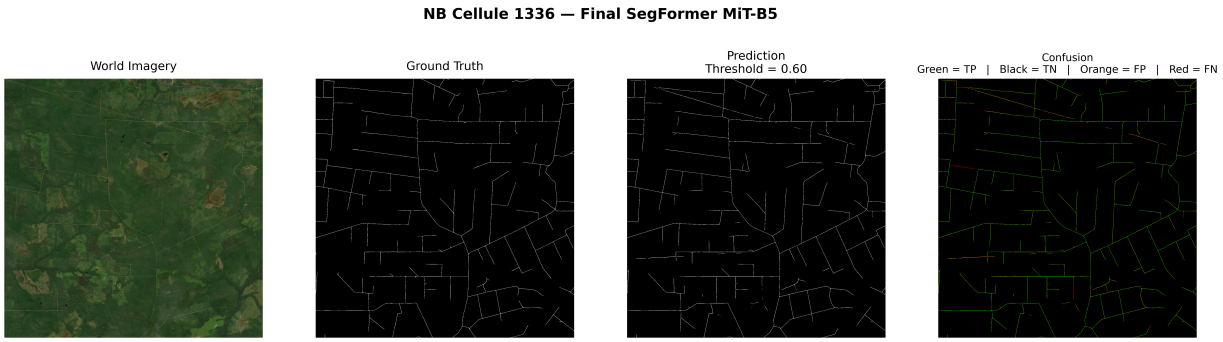
(a) Aggregate F1-score.

(b) Aggregate intersection over union.

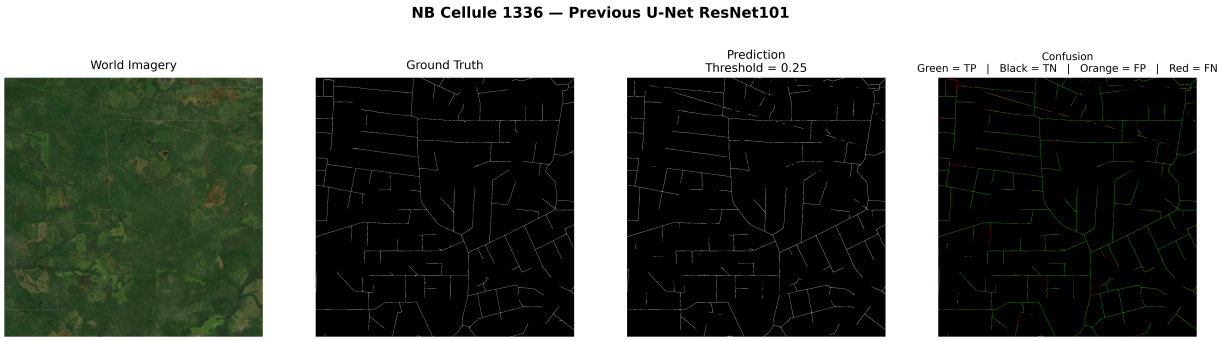
Figure 8: Threshold sensitivity across the six held-out validation sources using 50% overlap probability averaging.

5.4.3 Qualitative Validation Example

Figures 9 and 10 presents two examples of validation source for the final SegFormer and previous U-Net models. The figures contain the source World Imagery, corrected ground-truth mask, model prediction and pixel-wise confusion visualization. Green represents true positives, black true negatives, orange false positives and red false negatives. The displayed thresholds correspond to the best F1-Score threshold identified for each model on this validation source: 0.60 for SegFormer and 0.25 for the previous U-Net.

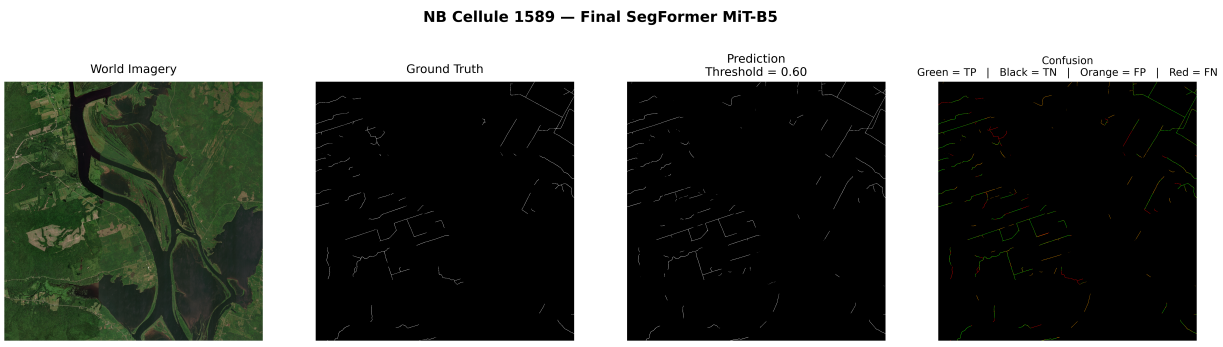


(a) Final SegFormer MiT-B5 at threshold 0.60.

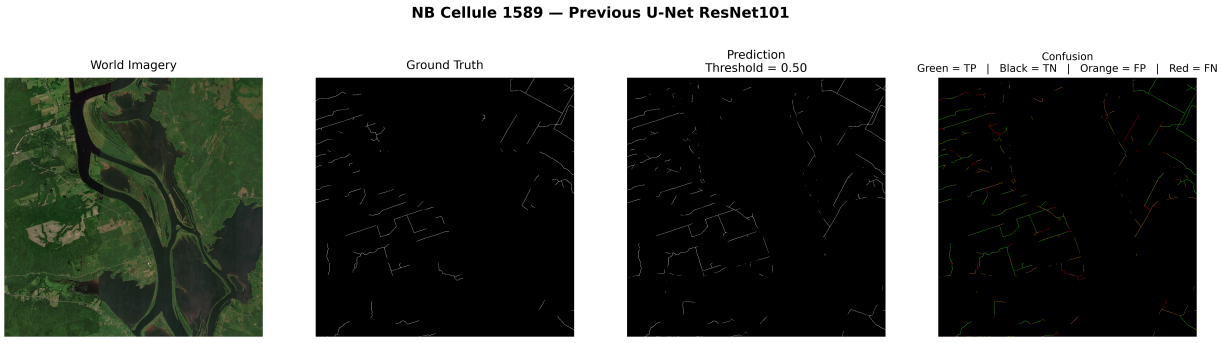


(b) Previous U-Net ResNet101 at threshold 0.25.

Figure 9: Qualitative comparison on the held-out `NB_Cellule_1336` source. From left to right: World Imagery, ground truth, binary prediction and confusion visualization.



(a) Final SegFormer MiT-B5 at threshold 0.60.



(b) Previous U-Net ResNet101 at threshold 0.55.

Figure 10: Qualitative comparison on the held-out `NB_Cellule_1336` source. From left to right: World Imagery, ground truth, binary prediction and confusion visualization.

These figures illustrate the differences in performance between the new SegFormer model and the previous U-Net model. Overall, the two models achieved relatively similar quantitative performance, indicating that both model architecture and dataset selection can have a comparable influence on segmentation performance. Several types of errors were also similar between the two models, such as the orange diagonal feature visible in both confusion matrices. In this case, both models produced a false positive according to the ground-truth annotation, even though a road is visibly present and could reasonably be classified as a road. SegFormer detected this road more consistently than U-Net, suggesting slightly better qualitative performance despite the similar quantitative metrics. A similar behavior can be observed in the second figure, where SegFormer produces more continuous, uniform, and spatially consistent road predictions than the older U-Net model, even though both models achieve comparable F_1-scores and other evaluation metrics. This highlights an important limitation of pixel-based evaluation for road segmentation, since exact pixel-level agreement with the annotations is not always required to obtain a useful representation of the road network. Visual analysis is therefore an important complement to quantitative metrics when evaluating the practical quality of the predictions. In some cases, road pixels may be missing or inaccurately represented in the ground-truth annotations, while the model correctly identifies the visible road, which can artificially reduce the reported performance metrics. This behavior is observed in several examples where the quantitative results of the two mod-

els are comparable, but SegFormer produces visually more consistent and complete road predictions.

Although SegFormer does not achieve perfect segmentation in every case, the majority of the road network is adequately detected, with only some artifacts, false detections, or missing road segments. Some of these errors can be reduced by adjusting the prediction threshold, allowing the final output to better represent the desired balance between road detection sensitivity and false-positive suppression. Threshold selection is therefore an important post-processing parameter for adapting the model output to the intended mapping application.

Overall, the combination of SegFormer and the revised dataset provides improved qualitative performance and more consistent road-network predictions, although some errors remain. Deep-learning segmentation models can still struggle in highly specific or complex situations, such as the second example, where the road network is discontinuous, narrow, irregular, or visually difficult to distinguish from the surrounding environment. Nevertheless, the model is able to adequately identify most of the visible roads in these challenging areas. False positives appear to be more common than false negatives, which can be preferable for this application because the primary objective is to identify and map as much of the road network as possible. In practice, detecting a potential road that can later be reviewed or filtered is generally less problematic than completely missing an existing road segment.

5.4.4 CPU–GPU and End-to-End Pipeline Timing

A separate timing experiment was performed on `NB_Cellule_1336` to compare the two deployment representations of the final SegFormer model with the previous U-Net CUDA model. For this benchmark, the complete World Imagery area was exported once as a single large raster and then reused for all three methods. Each method subsequently processed the same 169 overlapping windows. Separating the shared imagery-export time from model-specific processing makes it possible to distinguish GIS acquisition overhead from crop preparation, neural-network inference, overlap averaging and output-raster generation.

Table 9: End-to-end timing comparison on `NB_Cellule_1336`. The large World Imagery export was performed once and reused by all three methods.

Method	Export (s)	Load (s)	Read (s)	Crop/prep. (s)	Inference (s)	Overlap/avg. (s)	Write (s)	Map add (s)	Total (s)
SegFormer MiT-B5 – CUDA	48.36	5.60	0.79	4.74	35.81	0.41	6.12	0.46	104.00
SegFormer MiT-B5 – CPU	48.36	4.93	0.74	5.96	689.01	0.49	6.07	0.45	757.82
U-Net ResNet101 – CUDA	48.36	2.73	0.76	4.93	16.25	0.41	5.40	0.61	80.90

The shared large-image export required 48.36 s and was identical for all three evaluated configurations. The previous U-Net CUDA model required 16.25 s of ONNX inference time, the SegFormer CUDA model required 35.81 s, and the SegFormer CPU model required 689.01 s for the same 169 inference windows. When image export, model loading, image reading,

crop preparation, inference, overlap averaging, raster writing, and map insertion were combined, the total processing times were approximately 80.90 s for U-Net CUDA, 104.00 s for SegFormer CUDA, and 757.82 s for SegFormer CPU. This timing breakdown therefore separates the common GIS-related processing cost from the model-dependent computational cost under the same large-area inference workflow.

As expected, both CUDA-based configurations substantially outperformed CPU execution, reflecting the strong optimization of deep-learning workloads for GPU-based parallel computation. This difference demonstrates the importance of CUDA-capable GPUs for efficient deployment when large areas or large numbers of image tiles must be processed. Although the SegFormer CPU model remains technically usable and is provided as a deployment option, its inference time is considerably higher than that of the CUDA version. For the same 169 windows, SegFormer CPU inference required approximately 19 times more time than SegFormer CUDA inference, with 689.01 s compared with 35.81 s. The CPU version therefore provides an important compatibility option for systems without an NVIDIA GPU, but it is less suitable when rapid processing is required.

The performance difference between the two CUDA models is considerably smaller, although it remains noticeable. SegFormer required 35.81 s of inference time compared with 16.25 s for U-Net, corresponding to slightly more than twice the inference time. This increase is expected because SegFormer MiT-B5 is a larger and more computationally demanding architecture than the U-Net ResNet101 model. The additional computational cost is therefore a trade-off for the improved qualitative consistency and overall performance observed with SegFormer. Nevertheless, the U-Net model continues to provide reasonable segmentation results and may be preferable in scenarios where computational resources are limited or faster inference is prioritized over the additional performance provided by SegFormer.

Both SegFormer and U-Net will be made available in CUDA and CPU-compatible configurations within the GUI, providing a total of four deployment options. This allows the user to select the most appropriate model and execution mode according to the available hardware, required processing speed, and desired segmentation performance.

6 Conclusion

The results obtained throughout this project demonstrate that deep learning provides an effective and practical solution for large-scale forestry and agricultural road extraction from high-resolution remote sensing imagery. The first phase of the project established the feasibility of the proposed approach using a U-Net architecture with a ResNet101 encoder, which successfully detected the majority of forestry and agricultural roads while maintaining practical computational requirements for deployment. Quantitative and qualitative evaluation of this initial model nevertheless revealed several inconsistencies between the model predictions and the available ground-truth annotations. In particular, some paved roads and other visible road segments were not consistently represented in the annotations, causing visually

correct predictions to be penalized as false positives or false negatives. The Pixel Proximity Accuracy analysis further demonstrated that a considerable portion of the measured errors originated from small spatial offsets between predictions and annotations rather than complete failures to identify roads. These observations indicated that annotation quality and dataset consistency were important limiting factors in addition to model architecture.

Based on these findings, a second model-development phase was conducted using a revised and more consistent dataset. The corrected dataset was generated from 59 large source image–annotation pairs that were reviewed prior to training, with 53 source regions assigned to training and six reserved for validation using a source-level split to prevent spatial leakage. Road-aware sampling was then used to generate 21,200 training crops, while the six held-out validation sources produced 1,014 validation crops using 50% overlapping windows. This revised dataset was used to perform a broader model-selection study involving ten combinations of architecture and loss function, including U-Net, U-Net++, DeepLabV3+, MAnet and SegFormer. The results showed that most successfully trained architectures achieved relatively similar validation performance, with several configurations reaching an F_1-score close to 75%. This similarity indicates that the corrected dataset and improved sampling strategy played an important role in achieving stable road-segmentation performance and that increasing model complexity alone produced only modest improvements.

Among the evaluated configurations, SegFormer with a MiT-B5 encoder and Dice+Focal loss achieved the strongest overall validation results and was therefore selected as the final model. At its best operating threshold of 0.50, the selected SegFormer model achieved a Precision of 75.47%, Recall of 76.43%, F_1-score of 75.95% and IoU of 61.22%. Although several ResNet101-based architectures achieved very similar F_1-scores, SegFormer provided the highest overall F_1-score and IoU while maintaining a balanced Precision–Recall profile. These results suggest that the final model selection was not based on a large performance difference between architectures, but rather on SegFormer's small yet consistent advantage across the evaluated quantitative metrics and qualitative predictions.

Additional validation using the six held-out large source rasters confirmed this advantage under a deployment-style inference workflow. Using 50% overlapping 1024×1024 inference windows with probability averaging, SegFormer achieved a mean case-specific best F_1-score of 76.12% and a mean IoU of 61.70%, compared with 74.07% and 59.23%, respectively, for the previous U-Net model. SegFormer achieved a higher case-specific F_1-score and IoU on each of the six validation sources. When all six validation rasters were combined into a single pixel-weighted evaluation, SegFormer achieved an F_1-score of 76.83% and an IoU of 62.38%, compared with 75.32% and 60.41% for the previous U-Net. The qualitative analysis led to a similar conclusion, with SegFormer generally producing more continuous, complete and spatially consistent road predictions, even in cases where the quantitative differences between the two models were relatively small.

The final results also demonstrate that road-segmentation performance should be interpreted using both quantitative metrics and visual analysis. Roads are narrow linear structures,

and small spatial differences between predictions and annotations can substantially affect pixel-level metrics without necessarily reducing the practical usefulness of the extracted road network. Similarly, incomplete or inconsistent annotations can penalize predictions in locations where a road is visually present but not represented in the reference mask. Although the final SegFormer model does not perfectly segment every road, it adequately identifies the majority of visible road networks across the evaluated regions. The remaining errors generally occur in challenging situations involving narrow, discontinuous, visually ambiguous or complex road structures. Prediction-threshold adjustment also provides an additional mechanism for balancing missed road segments against false-positive detections according to the intended operational application. In this context, false positives may be more acceptable than false negatives because potential road detections can subsequently be reviewed or filtered, whereas completely missed road segments cannot be recovered without additional processing.

From a deployment perspective, the experiments demonstrate that both segmentation quality and computational efficiency must be considered when selecting a model. For the same 169 overlapping inference windows, the SegFormer CUDA model required 35.81 s of ONNX inference compared with 16.25 s for the U-Net CUDA model, while the SegFormer CPU model required 689.01 s. The corresponding end-to-end processing times were approximately 104.00 s for SegFormer CUDA, 80.90 s for U-Net CUDA and 757.82 s for SegFormer CPU. These results demonstrate the substantial advantage of CUDA acceleration for operational deployment, particularly when processing large geographic regions. The previous U-Net model therefore remains a valuable alternative when faster inference or lower computational requirements are prioritized. SegFormer, however, provides the strongest overall segmentation performance and was consequently retained as the primary final model.

Overall, the project progressed from an initial proof-of-concept model to a corrected dataset, a broader comparative model study and a final deployment-ready road-segmentation solution. The results demonstrate that data quality, annotation consistency and sampling strategy can influence segmentation performance as strongly as the choice of model architecture. The selected SegFormer MiT-B5 model provides the best overall balance of quantitative accuracy and qualitative road-network consistency, while the U-Net ResNet101 model remains a computationally efficient alternative. Together with the developed ArcGIS integration and ONNX deployment workflow, these models provide a scalable and practical framework for automated forestry and agricultural road mapping in New Brunswick and similar regions.

6.1 Future Directions and Next Steps

Although the final model and ArcGIS integration satisfy the objectives of the current project milestone and provide a complete operational deliverable, there remains considerable potential for future improvement. As with most artificial intelligence systems, segmentation performance can continue to improve as the quantity, diversity and quality of the available training data increase. One of the most important future directions would therefore be the

expansion of the training dataset to include a larger number of geographic regions, road environments and image acquisition conditions. A broader dataset would improve the model's ability to generalize to locations that differ from those represented during the current training process.

Future datasets could also incorporate temporal imagery acquired at different dates, seasons and years. Including seasonal variability would help the model better handle differences caused by vegetation growth, snow cover, shadows, soil conditions, road maintenance and other environmental changes that can substantially alter the visual appearance of roads. Similarly, incorporating imagery acquired at different spatial resolutions would improve robustness when the available source imagery does not exactly match the resolution used during training. A multi-temporal and multi-resolution dataset could therefore support a more general road-mapping model capable of operating consistently across a wider variety of remote-sensing conditions.

Another important extension would be to move beyond binary road-versus-background segmentation toward a multi-class road classification system. The current model focuses primarily on identifying the presence of roads, whereas future versions could distinguish between road categories such as paved roads, gravel roads, forestry roads, agricultural roads, trails and potentially abandoned or inactive road segments. Such a system could provide substantially more useful information for forestry management, transportation planning, environmental monitoring and road-inventory applications. However, this extension would require a substantial revision of the dataset because the existing annotations would need to be reviewed and relabeled using consistent road-class definitions. The corresponding models would also need to be retrained and re-evaluated specifically for multi-class segmentation.

Larger and more general deep-learning architectures could also be investigated as additional data become available. Foundation models and large pre-trained vision models are particularly promising because they can learn more general image representations and may adapt effectively to remote-sensing tasks using relatively limited task-specific fine-tuning. Such approaches could potentially improve robustness across different imagery sources, geographic regions and road types. However, these models generally require greater computational resources and more careful deployment optimization, meaning that any increase in segmentation performance would need to be balanced against inference speed, hardware requirements and operational complexity.

Additional work could also focus on improving the quality and consistency of the annotations themselves. The experiments conducted during this project demonstrated that annotation inconsistencies can directly affect measured model performance and can make visually correct predictions appear incorrect according to conventional pixel-level metrics. A more systematic annotation protocol, potentially including multiple reviewers or automated quality-control procedures, could therefore improve both model training and evaluation reliability. Future datasets could also include confidence levels or uncertain regions to better represent cases where the road boundary or road class is ambiguous.

Another useful research direction would be the development of active-learning or human-in-the-loop workflows. In such a system, the current model could first generate predictions over large geographic regions and human analysts could then review only uncertain or incorrect areas. These corrected predictions could subsequently be added back into the training dataset, allowing the model to improve progressively without requiring complete manual annotation of every new region. This approach could substantially reduce the cost of future dataset expansion while continuously adapting the model to new environments.

Temporal road-network monitoring could also extend the system beyond static road extraction. Repeated imagery from different years could be used to detect newly constructed roads, road extensions, road abandonment, vegetation regrowth or changes in road accessibility. This would transform the system from a road-mapping tool into a broader road-monitoring framework capable of identifying changes in forestry and rural transportation networks over time. Such functionality could be particularly valuable for environmental monitoring and for maintaining continuously updated road inventories.

From a deployment perspective, further optimization of the ArcGIS workflow could reduce processing time for very large study areas. The current experiments demonstrated that imagery export and GIS raster operations represent a substantial portion of the total execution time, particularly when GPU acceleration is used for neural-network inference. Future implementations could therefore investigate local imagery caching, direct raster access, more efficient parallel export strategies and additional batching or asynchronous processing methods. These improvements could reduce dependence on remote imagery services and further increase the practicality of province-scale processing.

Finally, future development could explore the use of uncertainty estimation to identify areas where model predictions are less reliable. Probability maps already provide useful information regarding model confidence, but more advanced uncertainty approaches could automatically highlight regions requiring manual review. Combining uncertainty estimation with active learning, multi-class segmentation and temporal monitoring could provide a more complete and adaptive road-mapping framework.

Overall, the current system provides a strong foundation for operational road extraction and successfully fulfills the objectives of the present project milestone. Future improvements are expected to depend primarily on expanding and diversifying the dataset, improving annotation consistency, incorporating temporal and multi-resolution imagery, extending the task toward multi-class road classification and evaluating larger pre-trained or foundation-model architectures. These developments would require additional dataset preparation, model training and validation, but they could progressively transform the current solution into a more general, robust and comprehensive road-mapping and monitoring platform.